

Shambhavi Mittal

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RESEARCH INTERESTS:

Network Measurement, Online Tracking, Usable Security, LLM security; I detect hidden mechanisms on digital platforms to understand and advocate about the risk to user privacy and security. I have used various methods, including web crawling, traceroute analysis, experiment set-up, latency comparisons, analyzing BGP routing, IP Geolocation, ping tests, multimodal AI evaluation, pattern recognition, and building custom crawlers. My goal is to develop methodologies and tools, while also informing policies to make digital platforms more transparent and privacy-preserving, helping users understand how their data is collected and utilized.

EDUCATION

University of California, San Diego

Master of Science, Computer Science

GPA - 3.9/4.0

Sep 2025 – Jun 2026

University of California, San Diego

Bachelor of Science, Computer Science

GPA - 4.0/4.0

Sep 2022 – Jun 2025

Latin honors: Summa Cum Laude (top 2%), Provost's Honors

WORK EXPERIENCE

- **Network Security Lab** | *Advised by: Dr. Alex Snoeren*

- **Research Assistant**

Jan 2025 – present

Identified 292 Google server-side trackers (270+ more than past work) across Tranco top 10K most popular websites by dissecting GTM using custom version of DuckDuckGo Tracker Radar Collector with manual server set-up and network request analysis. Engineered framework using cleaned data, anomaly detection, and validation checks; proposed generalizable methodology.

- **Student Researcher for Early Research Scholars Program**

Sep 2023 – Dec 2024

Analyzed internet traffic of 2000 government websites by measuring over 35k traceroutes to identify Country-in-the-Middle (CitM) cases where data traversed foreign countries unnecessarily. Performed IP path extraction, ping tests, and latency comparisons to detect routing detours.

- **Software Development Engineering Intern** at [Exaforce](#)

Jun 2025 – Sep 2025

Built and tested E2E automated SaaS database and graphics debugger for product pipeline, integrating agentic AI via Claude. Triggered from Slack through APIs and AWS lambda function launching Github workflow, dynamically deploying MCP servers (Jupyter and Superset). Reduced debugging time by 84.3% automating error detection.

- **Instructional Assistant for CSE 167: Computer Graphics** at [UCSD](#)

Sep 2024 – Mar 2025

Tutored over 410 students; Offered guidance on topics, assignments, projects and debugged C++ errors in office hours. Evaluated assignments for over 270 students; designed fair grading rubrics for 10 assignments.

- **AI & ML Fellow** at [Break Through Tech](#) by [Cornell](#) and [UCLA](#)

May 2024 – May 2025

Built 97.4% accurate knowledge graph for [BrightSide Health](#) applying LLM-based entity extraction to research papers, integrating results into queryable database and visualization to robustly identify treatments for depression and anxiety.

- **Software Engineering Intern** at [Ingenious Research Solutions](#)

Jul 2024 – Sep 2024

Supported development & deployment of [Divya Drishti](#), human recognition software with 95% accuracy, by implementing SDLC; Designed a ML model trained on 4000+ images to detect aircraft landing gear detection.

PUBLICATIONS

- **Shambhavi Mittal**, Alisha Ukani, A.C. Snoeren, “Hide and SST: Detecting and Analyzing Server-Side Tracking Across the Web” (PETS '25 Poster)

- Alisha Ukani, Katherine Izhikevich, **Shambhavi Mittal**, Manan Patel, Samvrit Srinath, Kristy Ly, KC Claffy, A.C. Snoeren, “Country-in-the-Middle: Measuring Paths between People and their Governments” (In Preparation, [ERSP '24 Poster](#), CNS '24 Poster)

PROJECTS

- **Noted** [[WebApp](#)] [[GitHub](#)] *Mar 2024 – Jun 2024*
Developed Noted, journal app for software devs following Software Development Life Cycle with 100% keyboard accessibility, comprehensive documentation, unit and E2E testing, automated CI/CD and calendar-based goal tracking.
- **Taxi Fares Prediction Model** [[GitHub](#)] *Sep 2024 – Dec 2024*
Spearheaded design and fine-tuning of neural network to predict NYC Taxi fares and tips with a RMSE of 0.377, 30x better than baseline models by carefully choosing features and processing data for CSE 158: Recommender Systems.
- **CAD Engineer** at [Center for Visual Computing](#) *Feb 2024 – Dec 2024*
Advised by [Dr. Ravi Ramamoorthi](#), Engineered a 3D device to reconstruct scene geometry to minimize specular reflections by 80% in Fusion 360.

SKILLS

- **Coding Languages:** Python, Java, C, C++, JavaScript, CSS, HTML, SQL, SQLite, Bash, Git
- **Technical tools and Skills:** agentic ingestion, data cleaning & processing, tool deployment, performance enhancement, distributed systems, visual studio, JDB, matplotlib, pandas, keras, LaTeX, OpenGL, Cursor, Claude, MCP Servers

AWARDS AND ACHIEVEMENTS

- Presented research poster at PETS '25 2025
- Awarded PETS Stipend to attend PETS '25 (USD 1000) 2025
- Presented research poster at and nominated to attend CNS Research Review 2024
- Presented research at Undergraduate Engineering Research Symposium | UCSD 2024
- Provost's Honors in every quarter of undergrad for academic excellence | UCSD 2022 to 2025
- Awarded Gold Medal for academic excellence for 7 consecutive years | DPS R.K. Puram 2021
- Awarded Blue Tie for receiving scholar badge for 6 consecutive years | DPS R.K. Puram 2020
- Awarded Gold Medal for academic excellence for 6 consecutive years | DPS Noida 2020

LEADERSHIP, OUTREACH, AND MENTORSHIP

- Vice President Finance at *Graduate Women in Computing (GradWIC)* *May 2025 – present*
- Beginner's Programming Competition Chair at *Women in Computing (WIC)* *Apr 2024 – Jun 2025*
Organized largest programming contest for 240+ coders; managed 12 people to make coding prompts in Java, C++, and Python.
- PR and Mentorship Director at *Computer Science and Engineering Society (CSES)* *Sep 2023 – Sep 2024*
- Outreach Mentor of *Eta Kappa Nu: Engineering Honor Society* *Sep 2023 – Sep 2024*
- EDGE Program Mentor of *WIC and Society of Women Engineers (SWE)* *Nov 2022 – Jun 2024*
- Founder and Student Mentor of [Stem Made Fun \(NGO\)](#) *Nov 2020 – present*